# Sheth L.U.J. And Sir M.V. College

Aim:Performance Evaluation of MANET Protocols
Measure throughput, packet delivery ratio, and end-to-end delay for different MANET routing protocols.

**CODE:**

```
# MANET_AODV.tcl

set val(chan)          Channel/WirelessChannel
set val(prop)          Propagation/TwoRayGround
set val(netif)         Phy/WirelessPhy
set val(mac)           Mac/802_11
set val(ifq)           Queue/DropTail/PriQueue
set val(ifqlen)        50
set val(ll)            LL
set val(ant)           Antenna/OmniAntenna
set val(nn)            5
set val(rp)            AODV
set val(x)             500
set val(y)             500
set val(stop)          20.0

set ns [new Simulator]

set tracefd [open manet.tr w]
$ns trace-all $tracefd

set namfd [open manet.nam w]
$ns namtrace-all-wireless $namfd $val(x) $val(y)

set topo [new Topography]
$topo load_flatgrid $val(x) $val(y)

create-god $val(nn)

set chan_1_ [new $val(chan)]

$ns node-config \
   -adhocRouting $val(rp) \
   -llType $val(ll) \
```

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```
    -macType $val(mac) \
    -ifqType $val(ifq) \
    -ifqLen $val(ifqlen) \
    -antType $val(ant) \
    -propType $val(prop) \
    -phyType $val(netif) \
    -channel $chan_1_ \
    -topoInstance $topo \
    -agentTrace ON \
    -routerTrace ON \
    -macTrace ON \
    -movementTrace ON

for {set i 0} {$i < $val(nn)} {incr i} {
    set node_(i)[ns node]
}

$node_(0) set X_ 50
$node_(0) set Y_ 100

$node_(1) set X_ 150
$node_(1) set Y_ 100

$node_(2) set X_ 250
$node_(2) set Y_ 100

$node_(3) set X_ 350
$node_(3) set Y_ 100

$node_(4) set X_ 450
$node_(4) set Y_ 100
for {set i 0} {$i < $val(nn)} {incr i} {
    node_(i) set Z_ 0
    $ns initial_node_pos node_(i) 20
}
# UDP/CBR Traffic
set udp0 [new Agent/UDP]
$ns attach-agent $node_(0) $udp0
set null0 [new Agent/Null]
$ns attach-agent $node_(4) $null0
$ns connect $udp0 $null0
```

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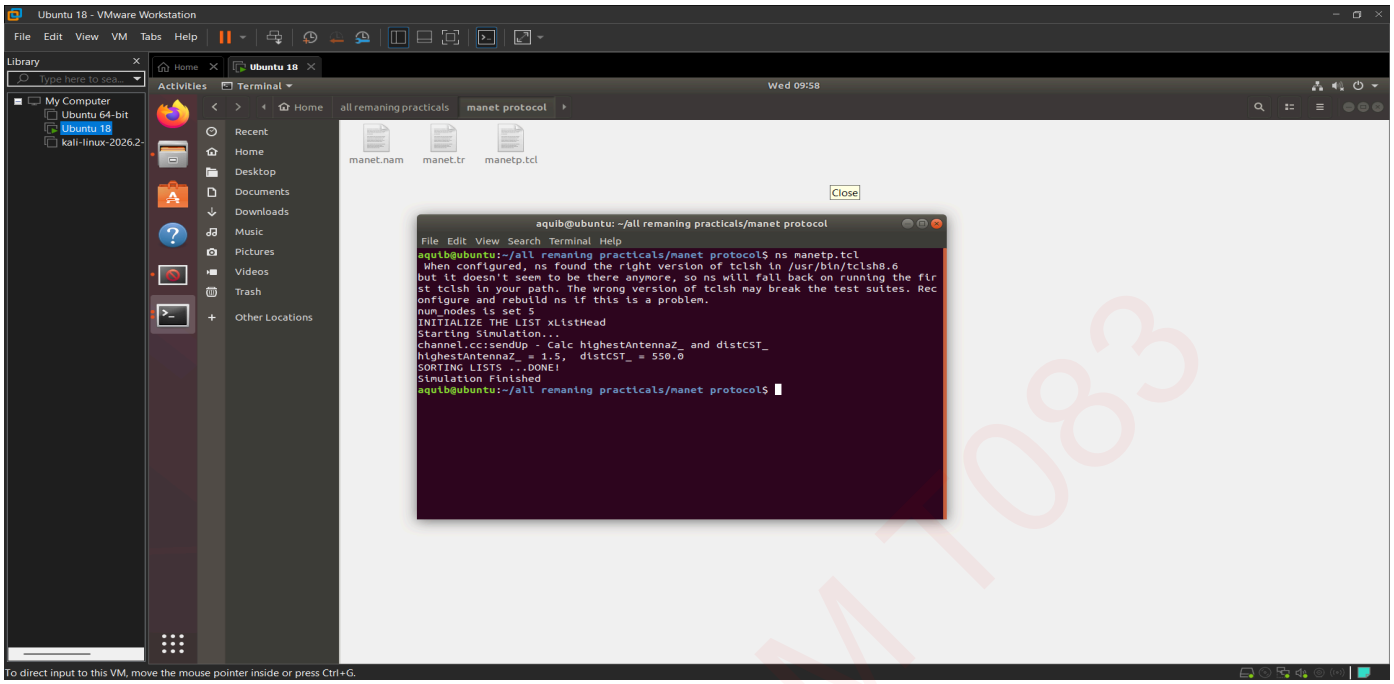
```
set cbr0 [new Application/Traffic/CBR]
$cbr0 set packetSize_ 512
$cbr0 set interval_ 0.05
$cbr0 attach-agent $udp0
nsat1.0"cbr0 start"
nsat18.0"cbr0 stop"
proc finish {} {
    global ns tracefd namfd
    $ns flush-trace
    close $tracefd
    close $namfd
    puts "Simulation Finished"
    exec nam manet.nam &
    exit 0
}
$ns at $val(stop) "finish"
puts "Starting Simulation..."
$ns run
```

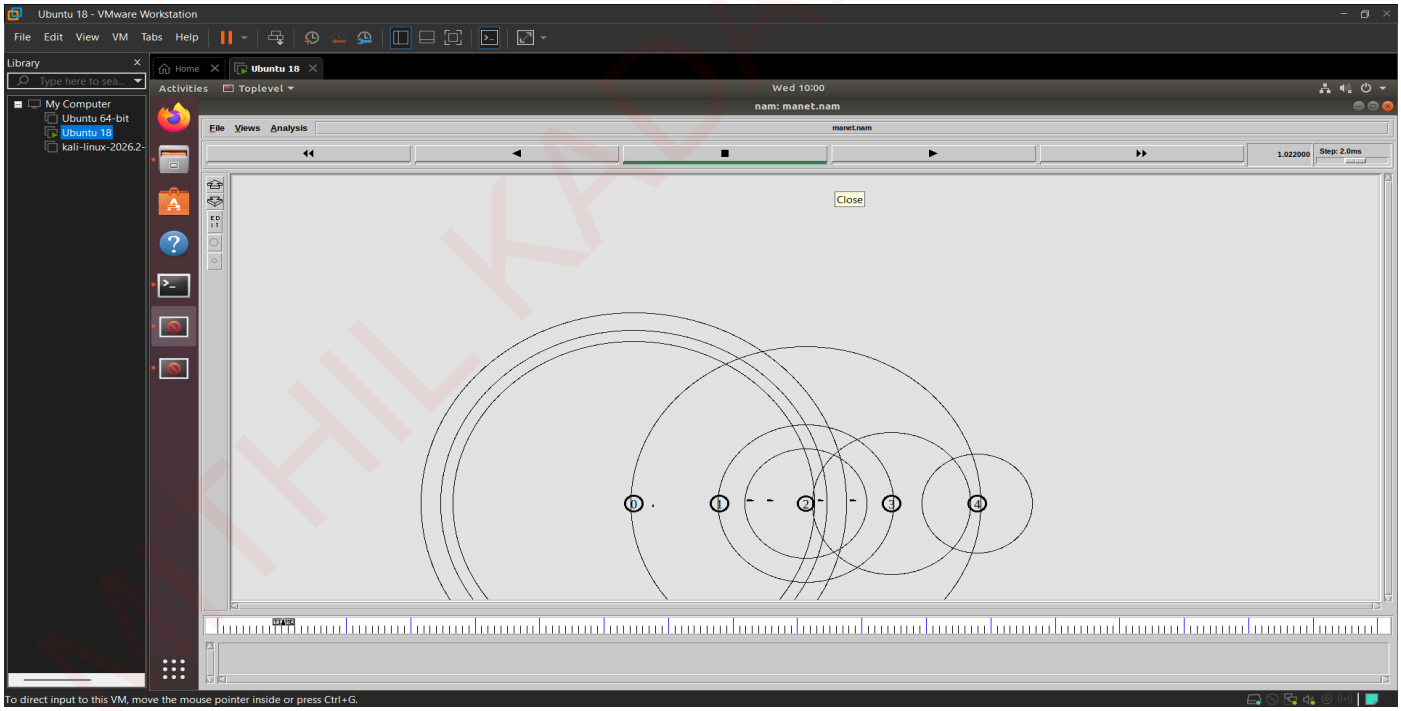
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OUTPUT:-

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